

Remote shell communication protocol specification v1.0.1

table of contents

[1. Physical layer](#)

[2. Data link layer](#)

[3. Network layer](#)

[4. Transport layer](#)

[5. Session layer](#)

5.1 Command list

5.2 Commands

[5.2.1 stty](#)

[5.2.2 \(blank line\)](#)

[5.2.3 HELO](#)

[5.2.4 put](#)

[5.2.5 send](#)

[5.2.6 get](#)

[5.2.7 delete](#)

[5.2.8 exec](#)

[5.2.9 reboot](#)

[5.2.10 ls](#) [5.2.11](#)

[dir](#)

[5.2.12 df](#)

[5.2.13 newfs](#)

[5.2.14 defrag](#)

[5.2.15 rename](#)

[5.2.16 speed](#)

[5.2.17 date](#)

[5.2.18 copy](#)

[5.2.19 setinfo](#)

[5.2.20 chmod](#)

[5.2.21 cd](#)

[5.2.22 pwd](#)

5.3 Text data transfer

5.4 File (binary data) transfer

[6. Presentation layer](#)

6.1 WonderWitch Transfer Format File Format

6.2 Response Text Data Format

6.2.1 stty

6.2.2 HELO

6.2.3 dir / ls

6.3 Prompt

1. Physical layer

RS232C

2. Data link layer

speed	38,400 / 9,600bps
data	8 bit
parity	none none
stop bit	1
Flow control	none
Sync	Synchronous pace (asynchronous)

3. Network layer

none

4. Transport layer

Stepless --command request, response
XMODEM --file transfer

5. Session layer

5.1 Command list

stty [<SP> <setting>] <line feed>
<line feed>
HELO [<SP> <greeting>] <line feed>
put [<SP> <absolute path directory name>] <line feed>
send [<SP> <absolute path Directory name>] <Line feed>
get <SP> <Absolute path file name> <Line feed>
get <SP> <File name> <Line feed>
delete <SP> <Absolute path file name> <Line feed>
delete <SP> <File Name> <Line feed>
exec <SP> <Absolute path file name> [<SP> <Parameter>] <Line feed>
exec <SP> <File name> [<SP> <Parameter>] <Line feed>
<Absolute path file name > [<Parameter>] <Line feed>
<File name> [<Parameter>] <Line feed>
reboot [<SP> <Number of seconds>] <Line feed>
ls [<SP> -l] [<SP> -d] [< SP> -ld] [<SP> <absolute path directory name>] <line feed>

```

dir [<SP> <absolute path directory name>] <line feed>
df [<SP> <file system name>] <line feed>
newfs < SP> <file system name> <line feed>
defrag <SP> <file system name> <line feed>
rename <SP> <absolute path file name> <SP> <new file name> <line feed>
rename <SP> <file name> <SP> <new file name> <line feed>
speed [<SP> <speed>] < Line feed>
date [<SP> <date and time> [<SP> <day>]] <line feed>
copy <SP> <source absolute path file name> <SP> <new absolute path directory name> <line feed>
copy <SP> <Source absolute path file name> <SP> <New absolute path file name> <Line feed> copy <SP> <Source file
name> <SP> <New absolute path directory name> <Line feed>
copy <SP> <File name> <SP> <New absolute path file name> <Line feed>
copy <SP> <Source absolute path file name> <SP> <Directory name> <Line feed>
copy <SP> <Source absolute path file name> <SP> < File name> <Line feed>
copy <SP> <File name> <SP> <Directory name> <Line feed>
copy <SP> <File name> <SP> <File name> <File line feed>
setinfo <SP> <Absolute path file name> <SP> <Description> <Line feed>
setinfo <SP> <File Name> <SP> <Description> <Line break>
chmod <SP> <Absolute path file name> <SP> <File mode> <Line break>
chmod <SP> <File name> <SP> <File mode> <Line break> <Line break> <Line break>
cd [<SP> <absolute path directory>] <line feed>
cd [<SP> <directory file name>] <line feed>
pwd <line feed>

cp reservation
rm reservation
mv reservation
move reservation
dirs reservation

```

[Syntax Rule]

<Absolute path directory name> ::= <File system directory> | <File system directory> <Directory>

<File system directory> ::= <Root file system directory> | <Root file system directory> <File system> | <Root file system directory> <File system>'/'

<Root file system directory> ::= '/'

<Directory> ::= <directory file> | <directory file>'/'

<Directory file> ::= <Subdirectory> | <Directory file> <Subdirectory>

<Subdirectory> ::= '/' <directory file name>

<Absolute path file name> ::= <File system directory> <File name> | <File system directory> <Directory> <File name>

<File system> ::= <Hard FS> | <Soft FS> | <Other FS>

<File system name> ::= <root FS> | <Hard FS> | <Soft FS> | <Other FS>

<root FS> ::= 'root'

<Hard FS> ::= 'rom' integer

<Soft FS> ::= 'ram' integer

<Other FS> :: = undefined

<Directory file name> :: = <file name>

<File name> :: String (other than path separator)

<Execution file name> :: Character string (except command character string)

<Settings> :: = String

<Greeting> :: = String

<Line break> :: = <CR>

<Description> :: = String

<Parameter> :: = Quart string | String

<Seconds> :: = Integer

<Integer> :: = Octal | Decimal | Hexadecimal

<Octal number> :: = 0 [0-9] *

<Decimal> :: = [1-9] +

<Hexadecimal> :: = 0x [0-9] +

<Day of the week> :: = [0-6]

5.2 Command

5.2.1 stty

stty [<SP> <setting>] <line feed>

Terminal mode settings.

<Setting> If there is no parameter, the current state is returned as text data.

When switching to interactive mode, a prompt is displayed after the response code * is sent *.

The default at startup is non-interactive mode

Parameters:

<Settings>

echo interactive mode (echo mode)

-echo non-interactive mode

Response code:

125 STARTING

(command accepted, transfer started)

451 LOCAL ERR

(command interrupted, local error)

502 NOT IMPLEMENTED

(command not implemented)
504 NOT IMPLEMENTED
(command parameters not implemented)
200 OK
(command succeeded)

5.2.2 (blank line)

<Line break>

Set terminal mode to interactive mode

When switching to interactive mode, a prompt is displayed after the response code * is sent *.
The default at startup is non-interactive mode

Response code:

451 LOCAL ERR
(command interrupted, local error)
502 NOT IMPLEMENTED
(command not implemented)
200 OK
(command succeeded)

5.2.3 HELO

HELO [<SP> <greeting>] <line feed>

Greet.

Return the reply greeting as text data.

Parameters:

<Greetings>

String-not processed

Response code:

125 STARTING
(command accepted, transfer started)
451 LOCAL ERR
(command interrupted, local error)
502 NOT IMPLEMENTED
(command not implemented)
200 OK
(command succeeded)

5.2.4 put

put [<SP> <absolute path directory name>] <line feed>

Send the file with XMODEM to the directory with the specified absolute path directory name. Do not confirm reception on the receiving side.

Parameters:

<Absolute path directory name>

Currently valid file system
rom0 --Hard File System # 0-> / rom0 | / rom0 /
ram0 --Soft File System # 0-> / ram0 | / ram0 /

* If omitted, it will be the current directory.

Response code:

350 FURTHER INFO
(command accepted, waiting for information)
426 TRANSFER ABOTED
(transfer interrupted)
450 NOT AVAILABLE
(file unavailable)
451 LOCAL ERR
(command interrupted, local error)
452 FS FULL
(File system full)
502 NOT IMPLEMENTED
(command not implemented)
504 NOT IMPLEMENTED
(command parameters not implemented)
550 NOT AVAILABLE
(file unavailable)
200 OK
(command successful)

5.2.5 send

send [<SP> <absolute path directory name>] <line feed>

Send the file with XMODEM to the directory with the specified absolute path directory name. Confirm reception on the receiving side.

Parameters:

<Absolute path directory name>

Currently valid file system
rom0 --Hard File System # 0-> / rom0 | / rom0 /
ram0 --Soft File System # 0-> / ram0 | / ram0 /

* If omitted, it will be the current directory.

Response code:

350 FURTHER INFO
(command accepted, waiting for information)
426 TRANSFER ABOTED
(transfer interrupted, reception rejected)
450 NOT AVAILABLE
(file unavailable)
451 LOCAL ERR
(command aborted, local) Error)
452 FS FULL
(file system full)
502 NOT IMPLEMENTED

(command not implemented)
504 NOT IMPLEMENTED
(command parameters not implemented)
550 NOT AVAILABLE
(file not available)
200 OK
(command successful) did)

5.2.6 get

get <SP> <absolute path file name> <line feed>
get <SP> <file name> <line feed>

Extract the file with the specified file name with XMODEM.

Parameters:

<Absolute path file name>

Currently valid file system

rom0 --Hard File System # 0

ram0 --Soft File System # 0

<File name>

File name of the current directory

Response code:

125 STARTING
(command accepted, transfer started)
450 NOT AVAILABLE
(file unavailable)
451 LOCAL ERR
(command aborted, local error)
452 FS FULL
(file system full)
502 NOT IMPLEMENTED
(command is Not implemented)
504 NOT IMPLEMENTED
(command parameters not implemented)
550 NOT AVAILABLE
(file not available)
553 NOT AVAILABLE
(file name not allowed)
200 OK
(command successful)

5.2.7 delete

delete <SP> <absolute path file name> <line feed>
delete <SP> <file name> <line feed>

Deletes the file with the specified file name.

Empty areas will not be collected until defragmented.

Parameters:

<Absolute path file name>

Currently valid file system

rom0 --Hard File System # 0

ram0 --Soft File System # 0

<File name>

File name of the current directory

Response code:

450 NOT AVAILABLE

(file not available)

451 LOCAL ERR

(command interrupted, local error)

502 NOT IMPLEMENTED

(command not implemented)

504 NOT IMPLEMENTED

(command parameters not implemented)

550 NOT AVAILABLE

(file)

553 NOT AVAILABLE

(filename not allowed) 200

OK

(command succeeded)

5.2.8 exec

(Format 1)

exec <SP> <absolute path file name> [<parameter>] <line feed>

exec <SP> <file name> [<parameter>] <line feed>

(Format 2)

<Absolute path file name> [<Parameter>] <Line feed>

<File name> [<Parameter>] <Line feed>

Executes the file with the specified file name.

(Search for commands on the file system first, and if it fails, search for internal commands)

Parameters:

<Absolute path file name>

Currently valid file system

rom0 --Hard File System # 0

<File name>

File name on the current directory

* When / ram0, exec fails

<Parameter>

A string or quote string.

If there are multiple parameters, separate them with a space.

Response code:

450 NOT AVAILABLE
(file not available)
451 LOCAL ERR
(command interrupted, local error)
502 NOT IMPLEMENTED
(command not implemented)
504 NOT IMPLEMENTED
(command parameters not implemented)
550 NOT AVAILABLE
(file)
553 NOT AVAILABLE
(filename not allowed) 200
OK
(command succeeded)

5.2.9 reboot

reboot [<SP> <integer>] <line feed>

Reboot after the specified number of seconds.
(Displays a reboot message up to the specified number of seconds)

Parameters:

<Seconds>

Seconds to reboot

* If omitted, 1 second

Response code:

451 LOCAL ERR
(command interrupted, local error)
502 NOT IMPLEMENTED
(command not implemented)
No reply on success. (To the opening)

5.2.10 ls

ls [<SP> -l] [<SP> -d] [<SP> -ld] [<SP> <absolute path directory name>] <line feed>

Returns a list of files on the specified file system as text data.

Parameters:

<Absolute path directory name>

Currently valid file system

rom0- Hard File System # 0-> / rom0 | / rom0 /

ram0-Soft File System # 0-> / ram0 | / ram0 / -root

File System

* If omitted, the current directory

Response code:

125 STARTING
(command accepted, transfer started)

451 LOCAL ERR
(command interrupted, local error)
502 NOT IMPLEMENTED
(command not implemented)
504 NOT IMPLEMENTED
(command parameters not implemented)
200 OK
(command succeeded)

5.2.11 dir

dir [<SP> <absolute path directory name>] <line feed>

Returns a list of files on the specified file system as text data.

Parameters:

<Absolute path directory name>

Currently valid file system

rom0 --Hard File System # 0

ram0 --Soft File System # 0

/ --root File System

* If omitted, the current directory

Response code:

125 STARTING
(command accepted, transfer started)
451 LOCAL ERR
(command interrupted, local error)
502 NOT IMPLEMENTED
(command not implemented)
504 NOT IMPLEMENTED
(command parameters not implemented)
200 OK
(command succeeded)

5.2.12 df

df [<SP> <filesystem name>] <line feed>

Returns a list of file system information as text data.

If a file system name is specified, only that file system

Parameters:

<File system name>

rom0 --Hard File System # 0

ram0 --Soft File System # 0

Response code:

125 STARTING
(command accepted, transfer started)
451 LOCAL ERR

(command interrupted, local error)
502 NOT IMPLEMENTED
(command not implemented)
200 OK
(command succeeded)

5.2.13 newfs

newfs <SP> <filesystem name> <line feed>

Initializes the specified file system.

Parameters:

<File system name>

rom0- Hard File System # 0

ram0- Soft File System # 0

Response code:

451 LOCAL ERR
(command interrupted, local error)
502 NOT IMPLEMENTED
(command not implemented)
504 NOT IMPLEMENTED
(command parameters not implemented)
200 OK
(command succeeded)

5.2.14 defrag

defrag <SP> <filesystem name> <line feed>

Defragment the specified file system.

Parameters:

<File system name>

rom0 --Hard File System # 0

ram0 --Soft File System # 0

Response code:

451 LOCAL ERR
(command interrupted, local error)
502 NOT IMPLEMENTED
(command not implemented)
504 NOT IMPLEMENTED
(command parameters not implemented)
200 OK
(command succeeded)

5.2.15 rename

rename <SP> <absolute path file name> <SP> <new file name> <line feed>
rename <SP> <file name> <SP> <new file name> <line feed>

Change the file name of the specified file name.

Parameters:

<Absolute path file name>

Valid file system

rom0- Hard File System # 0

ram0-Soft File System # 0

<File name>

File name on the current directory

<New file name>

String

Response code:

450 NOT AVAILABLE

(file unavailable)

451 LOCAL ERR

(command interrupted, local error)

502 NOT IMPLEMENTED

(command not implemented)

504 NOT IMPLEMENTED

(command parameters not implemented)

553 NOT AVAILABLE

(file) Name not allowed)

200 OK

(command succeeded)

5.2.16 speed

speed [<SP> <speed>] <line feed>

Set the communication speed.

If there is no <speed> parameter, the current state is returned as text data.

Parameters:

<Speed>

9600

38400

* If omitted, the current speed is returned.

Response code:

125 STARTING

(command accepted, transfer started)

451 LOCAL ERR

(command interrupted, local error)

502 NOT IMPLEMENTED

(command not implemented)

504 NOT IMPLEMENTED

(command parameters not implemented)

200 OK

(command succeeded)

5.2.17 date

date [<SP> <date time> [<SP> <day of the week>]] <line feed>

Set the RTC date and time.

If there is no <date time> parameter, the current state is returned as text data.

Parameters:

<Date time>

YYYYMMDDhhmm.ss

<Day of the week>

Numbers from 0 to 6

* 0 is Sunday, 6 is Saturday

* Added in 1.0.1 on Sunday

Response code:

125 STARTING

(command accepted, transfer started)

451 LOCAL ERR

(command interrupted, local error)

501 SYNTAX ERR

(syntax error in parameter)

502 NOT IMPLEMENTED

(command not implemented)

200 OK

(Command succeeded)

5.2.18 copy

copy <SP> <source absolute path file name> <SP> <absolute path directory name> <line feed>

copy <SP> <source absolute path file name> <SP> <absolute path file name> <line feed> copy <SP> <Source file name>

<SP> <Absolute path directory name> <Line feed>

copy <SP> <Source file name> <SP> <Absolute path file name> <Line feed>

copy <SP> <Source absolute path file name> < SP> <directory name> <line feed>

copy <SP> <source absolute path file name> <SP> <file name> <line feed>

copy <SP> <source file name> <SP> <directory name> <line feed>

copy <SP> <source file name> <SP> <file name> <line feed>

Copy the specified file.

Parameters:

<Source absolute path file name>

<File name>

<Absolute path directory name>

<Directory name>

Valid file system name

rom0 --Hard File System # 0

Response code:

450 NOT AVAILABLE
(file unavailable)
451 LOCAL ERR
(command interrupted, local error)
502 NOT IMPLEMENTED
(command not implemented)
504 NOT IMPLEMENTED
(command parameters not implemented)
553 NOT AVAILABLE
(file) Name not allowed)
200 OK
(command succeeded)

5.2.19 setinfo

setinfo <SP> <absolute path file name> <SP> <description> <line feed>
setinfo <SP> <file name> <SP> <description> <line feed>

Change the description of the specified file name.
(Maximum 24 bytes. Allow space)

Parameters:

<Absolute path file name>

Valid file system

rom0 --Hard File System # 0

ram0 --Soft File System # 0

<File name>

File name on the current directory

<Explanation>

String or quote string

Response code:

450 NOT AVAILABLE
(file unavailable)
451 LOCAL ERR
(command interrupted, local error)
502 NOT IMPLEMENTED
(command not implemented)
504 NOT IMPLEMENTED
(command parameters not implemented)
553 NOT AVAILABLE
(file) Name not allowed)
200 OK
(command succeeded)

5.2.20 chmod

chmod <SP> <absolute path file name> <SP> <file mode> <line feed>
chmod <SP> <file name> <SP> <file mode> <line feed>

Change the file mode of the specified absolute path file name.

Parameters:

<Absolute path file name>

Valid file system

rom0 --Hard File System # 0

ram0 --Soft File System # 0

<File name>

File name on the current directory

<File mode>

Octal (numbers starting from 0) or

hexadecimal numbers (numbers starting from non-zero)

Response code:

450 NOT AVAILABLE

(file unavailable)

451 LOCAL ERR

(command interrupted, local error)

502 NOT IMPLEMENTED

(command not implemented)

504 NOT IMPLEMENTED

(command parameters not implemented)

553 NOT AVAILABLE

(file) Name not allowed)

200 OK

(command succeeded)

5.2.21 cd

cd [<SP> <absolute path directory>] <line feed>

cd [<SP> <directory name>] <line feed>

Change the current directory to the specified directory

Parameters:

<Absolute path directory name>

Valid file system

rom0 --Hard File System # 0

ram0 --Soft File System # 0

<Directory name> Directory name

from the current directory

Response code:

450 NOT AVAILABLE

(file unavailable)

451 LOCAL ERR
(command interrupted, local error)
502 NOT IMPLEMENTED
(command not implemented)
504 NOT IMPLEMENTED
(command parameters not implemented)
553 NOT AVAILABLE
(file) Name not allowed)
200 OK
(command succeeded)

5.2.22 pwd

pwd <line feed>

Returns the current directory

Response code:

125 STARTING
(command accepted, transfer started)
451 LOCAL ERR
(command interrupted, local error)
501 SYNTAX ERR
(syntax error in parameter)
502 NOT IMPLEMENTED
(command not implemented)
200 OK
(Command succeeded)

5.3 Text data transfer

The transfer of text data is realized as text sandwiched between the .

The'!' At the beginning of the line of the text data itself is escaped to'..'and sent.
The receiver needs to decode this to'!'.

example:

[Protocol data]

```
meg> HELO <line feed> <-command (send on request side)
125 STARTING <line feed> <-continuous response
HELO <line feed> <-text data
WonderWitch 1.0.0 <line feed> <-text data
FreyaOS 1.0.0 <line feed> < --Text data
FreyaBIOS 1.0.0 <Line feed >> <-Text data
c / w Meg xxxx <Line feed> <-Text data
Owner xxxx <Line feed> <-Text data
. <Line feed> <-EOT
200 OK <Line feed> < -Result response
```

[Text data]

```
HELO <line feed>
WonderWitch 1.0.0 <line feed>
FreyaOS 1.0.0 <line feed>
FreyaBIOS 1.0.0 <line feed>
c / w Meg xxxx <line feed>
Owner xxxx <line feed>
```


5.4 File (binary data) transfer

File transfer is performed by XMODEM transfer after the response in the 100s (at the time of reception) or 300s (at the time of transmission), which is a continuous response .

For the format of the file transferred by XMODEM, refer to "6.1 File transfer format file format".

Example 1:

[Protocol data]

```
meg> put / rom0 <line feed> <-command (send on request side)
350 FURTHER INFO <line feed> <-continuous response
(XMODEM reception) <-binary data
200 OK <line feed> <-result response
```

Example 2:

[Protocol data]

```
meg> get / rom0 / SomeFile <line feed> <-command (send on request side)
125 STARTING <line feed> <-continuous response
(XMODEM transmission) <-binary data
200 OK <line feed> <-result response
```

Example 3:

[Protocol data]

```
meg> send / rom0 <line feed> <-command (request side send)
426 TRANSFER ABOTED <line feed> <-receiver reject
```

6. Presentation layer

6.1 WonderWitch Transfer Format File Format

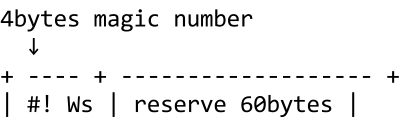
The first 128 bytes of the WonderWitch transfer format file stores the file's common header information and is used as the first XMODEM transfer block (block # 0).

After that, the file body data is stored.

WonderWitch transfer format file:

Location	content
0	--3 Magic number "#! Ws" (0x21 0x23 0x73 0x77 in hexadecimal)
4	--63 reserve (usually padded with FFH)
64	--127 File information (<sys / filesys.h> The content of the defined fent_t itself)
128	--EOF file body (including resources, if any)

Block when transferring files



```

+ ---- + |
| (usually FF padding ) |
+ ----- + block # 0
| | (128bytes)
| File information 64bytes |
| |
+ ----- +

+ ----- +
| |
| |
| File body 128bytes | block # 1
| |
+ ----- +

+ ----- +
| |
| |
| File body 128bytes | block # 2
| |
+ ----- +

+ ----- +
| |
| |
| File body 128bytes | block # 2
| |
+ ----- +

:::

```

6.2 Response text data format

6.2.1 stty

```

<echo on time>

echo echo

<at echo off>

-echo

```

6.2.2 HELO

String

6.2.3 dir / ls

```

total xx files
filename drwx = size time description

d is one of the following three d --directory
file (archive file)
i --indirect library
l --symbolic link

r --readable
w --writable
x --executable

= --stream accessible flag (read data)

```

size is decimal right justified ULONGMAX (unsigned long)
time is a string YYYY-MM-DD hh: mm: ss (see date command)

6.2.4 df

```
Filesystem Size Used Available Capacity Entries
root 64 64 0 100% 4
rom0 327680 116992 210688 35% 128
ram0 65536 12160 53376 18% 64
```

(avail = size --used)

6.2.5 date

YYYY-MM-DD hh: mm: ss w

* W (day of the week) is added in 1.0.1

6.3 Prompt

Shell is "meg>" in Meg

reference:

--FTP RFC Japanese version

<http://hp.vector.co.jp/authors/VA002682/rfc959j.htm>

--XMODEM

<ftp://ftp.std.com/obi/Standards/FileTransfer/YMODEM8.DOC.1.Z>

<http://www.st.rim.or.jp/~phinloda/proto2.html>